The Refractory Period

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

Information text

The refractory period is the time after an action potential when a membrane section cannot be excited again, or only with difficulty. During depolarisation, voltage-gated sodium channels open and the membrane potential becomes more positive. During repolarisation, sodium channels are inactivated and potassium channels let positive charge leave the cell. In the absolute refractory period, no new action potential can be triggered. In the relative refractory period, excitability returns gradually, but a stronger stimulus is needed.

Task 1: Absolute and relative refractory period

Explain the difference between the absolute and relative refractory period using the states of the voltage-gated sodium channels.

Task 2: Importance of the refractory period

Create a hypothesis about the importance of the refractory period for directed conduction and maximum signal frequency.

Task 3: Observation in the model

Describe excitability in the blue-marked areas. Focus on voltage-gated sodium channels.